

Jenkins Slave Configuration

TCP Connection- over TCP port

Two methods to achieve Jenkins slave configuration:

1. **SSH** - Port 22
2. **JNLP** - Java Network Launch Protocol - with any random TCP port (0 - 65535)

SLAVE / NODE / AGENT / LABEL - all are same

Dashboard → Manage Jenkins → Nodes

You can see all your nodes here,
Jenkins master - [Build-In Node](#)

Let's launch and prepare a slave machine: (slave-1)

Launch an Ec2 instance → name (slave-1) → login to machine →

```
$ sudo su -
```

```
# amazon-linux-extras install java-openjdk11 -y
```

(install java on slave machine *must have java on slave machine)

Create one directory as a slave's remote root directory where we will download our agent.jar file later in this practical.

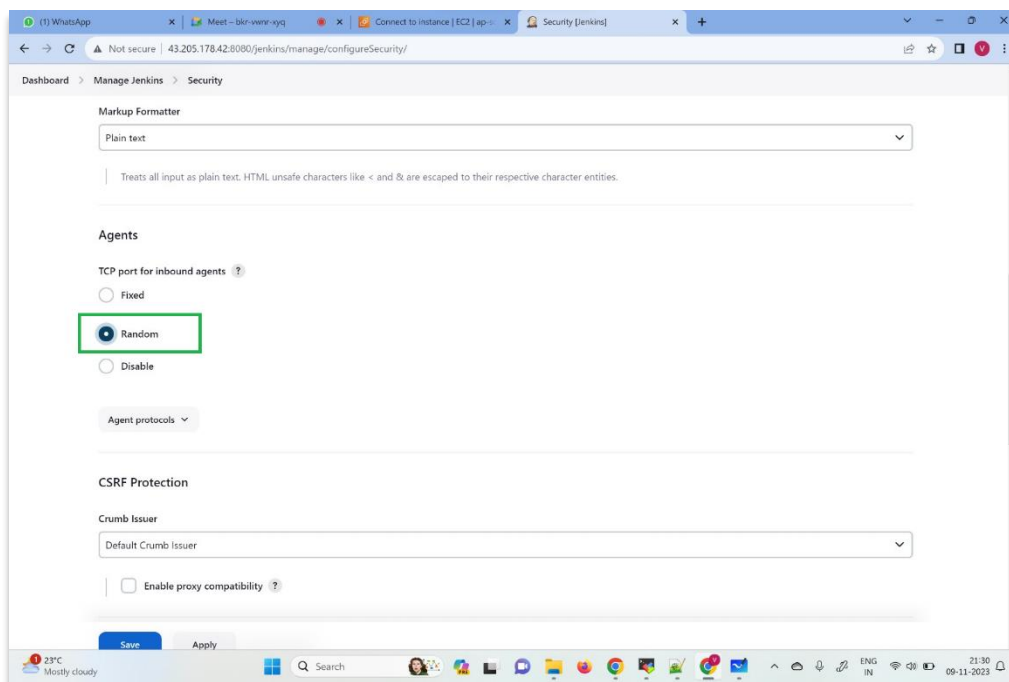
Eg:

```
cd /mnt
```

```
mkdir jenkins-slave
```

JNLP Method:

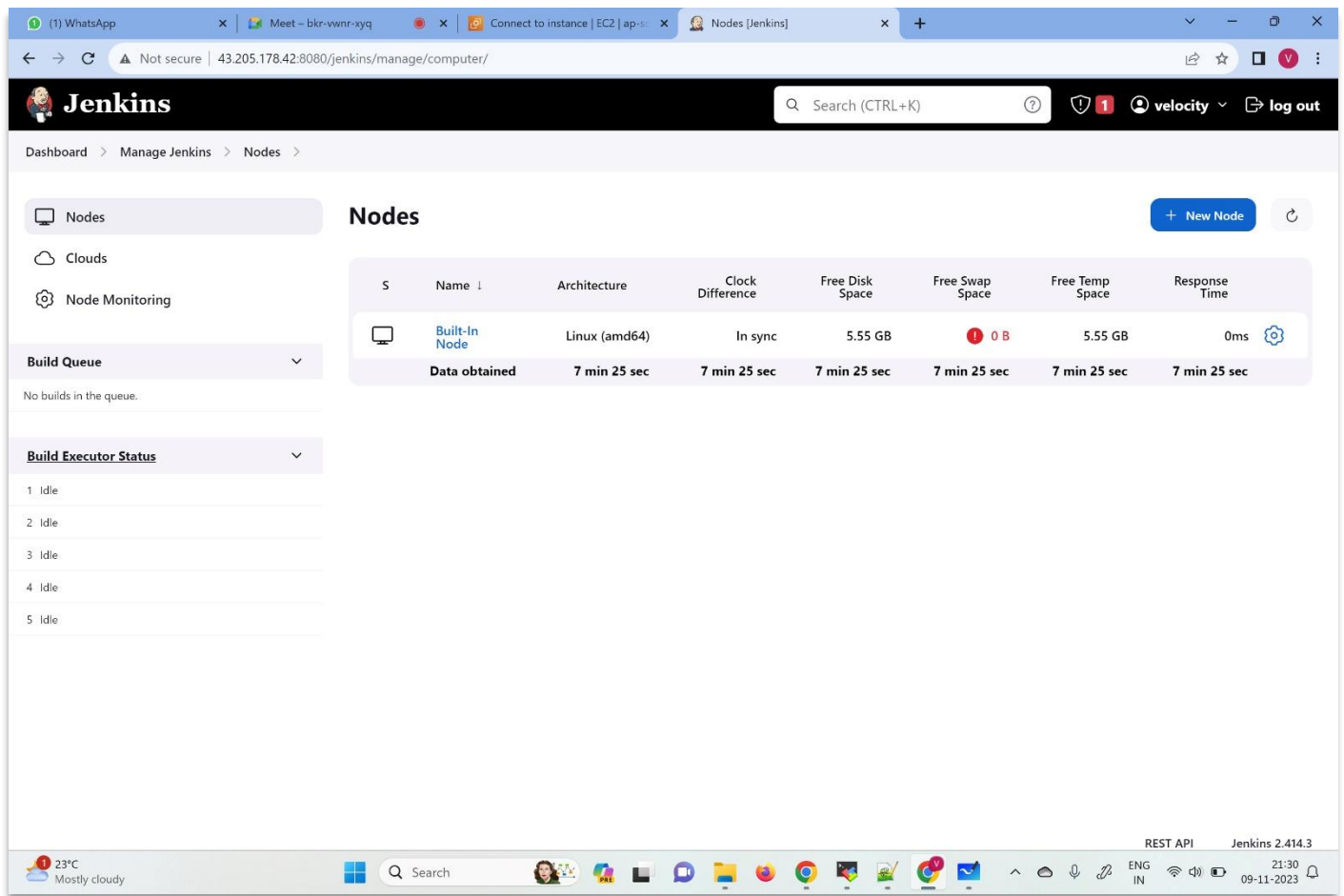
Go to Manage Jenkins → Security → Agents → select TCP port for inbound agents as **Random** (we will make it **Fixed** later)



→ **Apply + Save**

We will create a slave now

Dashboard → Manage Jenkins → Nodes →



The screenshot shows the Jenkins web interface. The top navigation bar includes the Jenkins logo, a search bar, and user information. The main content area is titled 'Nodes' and features a table of node details. The table has columns for 'S', 'Name', 'Architecture', 'Clock Difference', 'Free Disk Space', 'Free Swap Space', 'Free Temp Space', and 'Response Time'. The first row shows a 'Built-In Node' with 'Linux (amd64)' architecture, 'In sync' clock difference, and '5.55 GB' free disk space. Below the table, there is a 'Build Queue' section with a dropdown menu and a list of build executors, all showing 'Idle' status.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	5.55 GB	0 B	5.55 GB	0ms
	Data obtained	7 min 25 sec	7 min 25 sec	7 min 25 sec	7 min 25 sec	7 min 25 sec	7 min 25 sec

Click on New node → add node name (slave-1) → select Type Permanent Agent → click Create → add description → Number of executors (eg: 5) → add Remote root directory (Eg: /mnt/jenkins-slave ; this directory has to be present on machine Jenkins will not create for us) → add label (must have, eg: dev) → Launch method → Launch agent by connecting it to the connector (this is JNLP) → click on **save**

(1) WhatsApp x Meet - bkr-vwnr-xyq x Instances | EC2 | ap-south-1 x Jenkins x +

Not secure | 43.205.178.42:8080/jenkins/manage/computer/createItem

Dashboard > Manage Jenkins > Nodes >

Name ?

slave-1

Description ?

this is slave-1

Plain text Preview

Number of executors ?

5

Remote root directory ?

/mnt/jenkins-slave

Labels ?

dev

Usage ?

Use this node as much as possible

Launch method ?

Save

23°C Mostly cloudy 21:33 09-11-2023

(1) WhatsApp x Meet - bkr-vwnr-xyq x Instances | EC2 | ap-south-1 x Jenkins x +

Not secure | 43.205.178.42:8080/jenkins/manage/computer/createItem

Dashboard > Manage Jenkins > Nodes >

Usage ?

Use this node as much as possible

Launch method ?

Launch agent by connecting it to the controller

☐ Disable WorkDir ?

Custom WorkDir path ?

Internal data directory ?

remoting

☐ Fail if workspace is missing ?

☐ Use WebSocket ?

Advanced

Availability ?

Keep this agent online as much as possible

Save

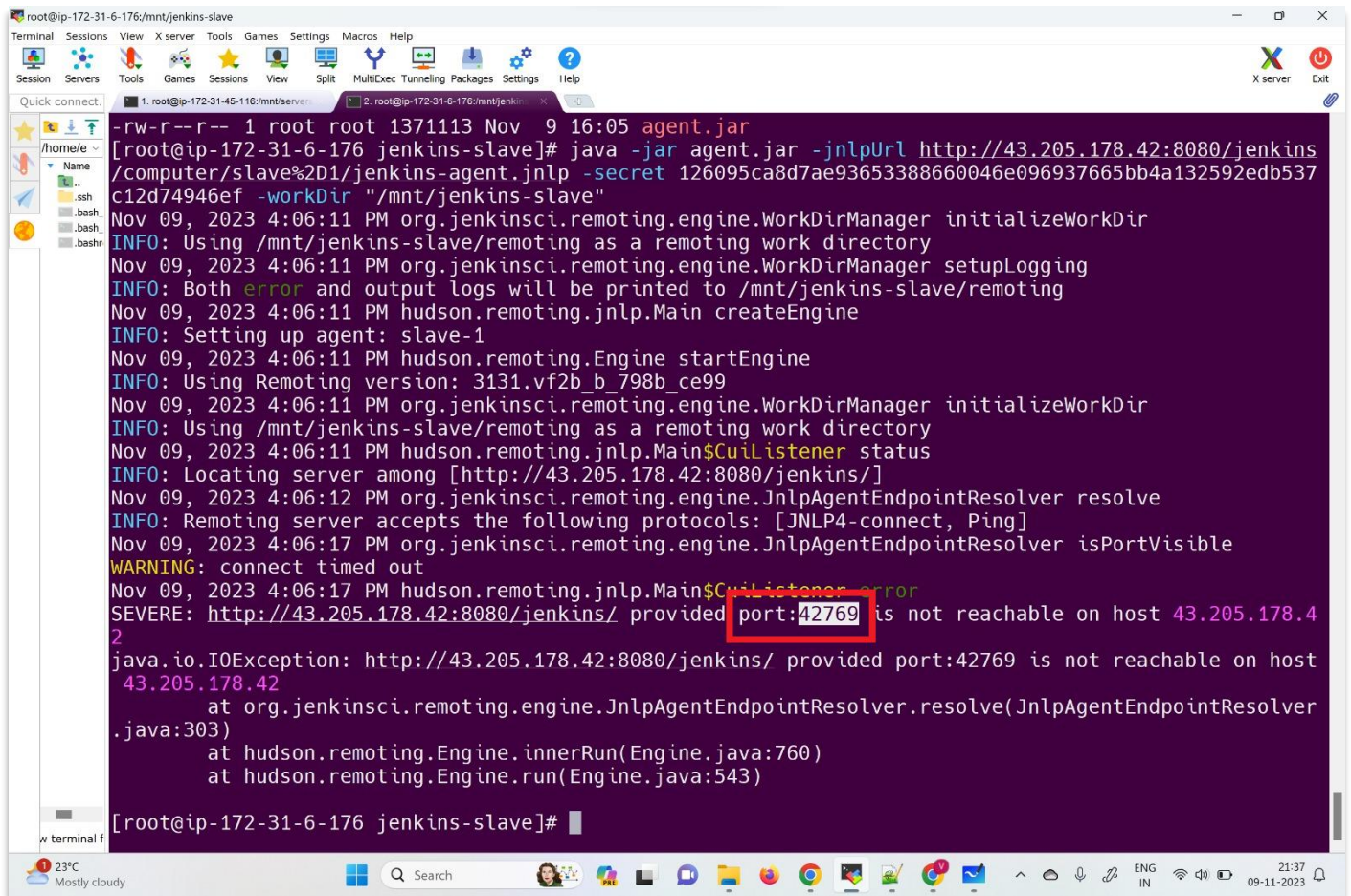
23°C Mostly cloudy 21:34 09-11-2023

You can see new slave-1 is created but it's disconnected, now open slave-1 node you will see a few commands. We will connect our slave to master with the help of this commands: (check if Jenkins URL is updated by going to - Dashboard – Manage Jenkins – System – Jenkins URL)

Now download agent.jar on slave machine, inside slave's Remote root directory (Eg: /mnt/jenkins-slave) using curl command.

Again, run the second command starting with - java -jar

It will try to connect but will give an error, in the logs we can see why it failed to connect over a specific port number. The reason is that the port number is not open in our machine's security group from where it will attempt to connect with our Master. we can also add this port number in Jenkins security and make it fixed.



```
root@ip-172-31-6-176:/mnt/jenkins-slave
-rw-r--r-- 1 root root 1371113 Nov  9 16:05 agent.jar
[root@ip-172-31-6-176 jenkins-slave]# java -jar agent.jar -jnlpUrl http://43.205.178.42:8080/jenkins/computer/slave%2D1/jenkins-agent.jnlp -secret 126095ca8d7ae93653388660046e096937665bb4a132592edb537c12d74946ef -workDir "/mnt/jenkins-slave"
Nov 09, 2023 4:06:11 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /mnt/jenkins-slave/remoting as a remoting work directory
Nov 09, 2023 4:06:11 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /mnt/jenkins-slave/remoting
Nov 09, 2023 4:06:11 PM hudson.remoting.jnlp.Main createEngine
INFO: Setting up agent: slave-1
Nov 09, 2023 4:06:11 PM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3131.vf2b_b_798b_ce99
Nov 09, 2023 4:06:11 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /mnt/jenkins-slave/remoting as a remoting work directory
Nov 09, 2023 4:06:11 PM hudson.remoting.jnlp.Main$CuiListener status
INFO: Locating server among [http://43.205.178.42:8080/jenkins/]
Nov 09, 2023 4:06:12 PM org.jenkinsci.remoting.engine.JnlpAgentEndpointResolver resolve
INFO: Remoting server accepts the following protocols: [JNLP4-connect, Ping]
Nov 09, 2023 4:06:17 PM org.jenkinsci.remoting.engine.JnlpAgentEndpointResolver isVisible
WARNING: connect timed out
Nov 09, 2023 4:06:17 PM hudson.remoting.jnlp.Main$CuiListener error
SEVERE: http://43.205.178.42:8080/jenkins/ provided port:42769 is not reachable on host 43.205.178.42
java.io.IOException: http://43.205.178.42:8080/jenkins/ provided port:42769 is not reachable on host 43.205.178.42
    at org.jenkinsci.remoting.engine.JnlpAgentEndpointResolver.resolve(JnlpAgentEndpointResolver.java:303)
    at hudson.remoting.Engine.innerRun(Engine.java:760)
    at hudson.remoting.Engine.run(Engine.java:543)
[root@ip-172-31-6-176 jenkins-slave]#
```

For that go to, Dashboard → Manage Jenkins → Security → Inside Agents → tcp port for inbound → select Fixed and add port number here.

We need to open the same port on the machine's security group then run the same command again and we can see that our slave will now connect to our Master.

Check by going to → Dashboard → Manage Jenkins → Nodes → slave-1

If we press CTRL+C on terminal it will lose the connection with master, hence to run the command in background Use same command starting with 'nohup' and ending with '&'

Eg: # nohup <command> &

Run the command now and press enter key 2 - 3 times. our command is running in the background.

Command History of Slave Machine

```
# amazon-linux-extras install java-openjdk11
# cd /mnt
# mkdir jenkins-slave
# cd jenkins-slave/
# curl -sO http://43.205.178.42:8080/jenkins/jnlpJars/agent.jar
# nohup java -jar agent.jar -jnlpUrl
http://43.205.178.42:8080/jenkins/computer/slave%2D2/jenkins-agent.jnlp
-secret
07fea906afed6aaca3d88a5338c67b6d50b9bc31eb40e4d13908592aaa658cf8 -
workDir "/mnt/jenkins-slave" &
tail -100f c
tail -100f nohup.out
```

Now, create one more machine (slave-2) by repeating above steps.

Lets run a job:

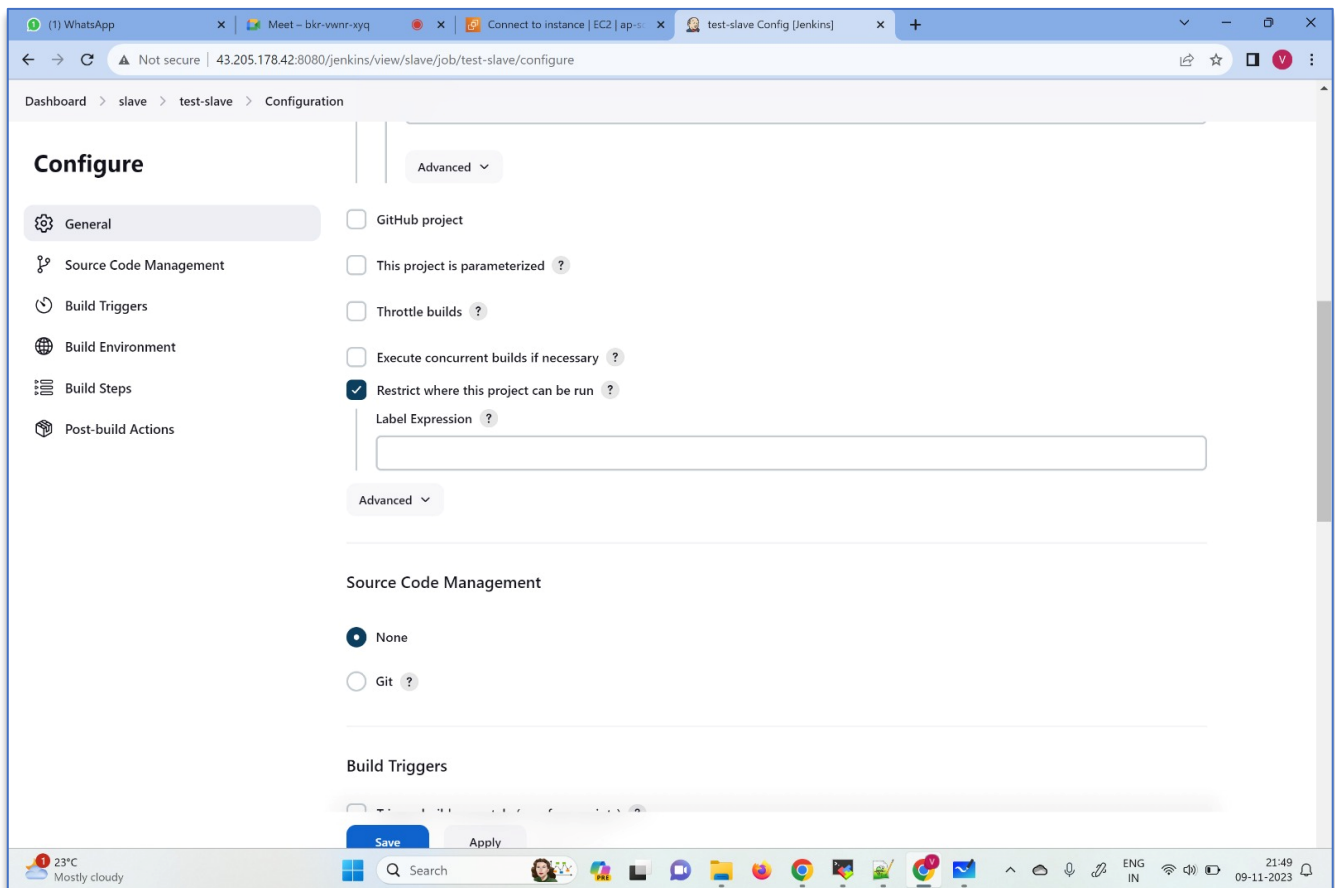
New Item → add name (test-slave) → Freestyle Project → OK → Add build steps → execute shell

```
echo "hello!!!"
```

→ select Delete workspace before build start → Discard Old builds → **Apply + Save**

We haven't mentioned anything about slaves related to this job so by default Jenkins will run jobs on any available free slaves.

To specify where our job should run:



Open Job and go to configure → select restrict where this project can be run → mention Label Expression > add 'dev' (it will run job on any free machine which has label dev) → **Apply + Save**

Let's say if we want to run job on particular slave - just write name of the slave instead of label (Eg: slave-1) next time when you build a job it will execute on that particular machine.